

JUAN DIEGO MENDEZ

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EDUCATION

The University of Texas at Austin, Austin, TX
Bachelor of Science in Computer Science

Jan 2022 – May 2025

SKILLS

Programming Languages: C#, Python, C, C++, JavaScript, TypeScript, Java, HTML, CSS, SQL.

Tools: Unity, VS Code, Scikit-learn, TensorFlow, Blender, WebGL, Docker, Git, Bitbucket, Node.js, Linux, OpenCV

Relevant Coursework and education: Data Structures · Discrete Math · Computer Architecture · Intro to Probability and Statistics · Principles of Computer Systems (OS) · Matrices and Matrix Calculations · Cyber-physical systems · Game Technology · Programming for Performance · Computer Graphics · Algorithms & Complexity · Computer Vision

EXPERIENCE

Longhorn Neurotech, Austin, TX

Sep 2024 – May 2025

Team Lead for VR Design

- **Unity 3D & C# Programming:** Led a team to design and implement a fully interactive VR environment in Unity 3D using C#, tailored for integration with an EEG-based BCI system.
- **Signal Integration & Input Controller Design:** Led team to create an Input Controller in Unity to receive and map preprocessed EEG signals into game actions, enabling real-time neuro-interaction within the VR environment.
- **Cross-Team Coordination:** Collaborated with signal processing and ML subteams to ensure seamless interfacing between the EEG data pipeline and VR environment, facilitating synchronized real-time feedback.

Monkeyflux S.L., Madrid, Spain (remote)

Jun 2023 – Aug 2023

Software Engineer Intern

- **Front-End Development:** Enhanced the UI and functionality of MonkeyFlux Babuin's web platform, refining user experience and improving site performance. Implemented new features, addressed bugs, and optimized existing code using TypeScript, HTML, CSS.
- **DevOps & Environment Setup:** Worked with Docker containers and WSL to configure a local development environment for seamless collaboration

PROJECTS

Research on Mixed Reality Environment Interactions using a Depth Camera, Austin, TX

Jan 2024 – May 2024

Principal Investigator: Developed a system that allows real-time interactions between virtual objects and physical space

- **Virtual Reality & Computer Vision:** Developed a system enabling real-time interaction between virtual and physical objects using Unity, C#, and OpenCV.
- **Hardware Integration:** Integrated Oculus Quest 2 and Intel RealSense D435 for depth sensing and spatial awareness.
- **Optimization Techniques:** Utilized voxel-based approaches to accelerate computation and improve system efficiency.

EEG-Based Reinforcement Learning for Binaural Beat Optimization, Austin, TX

Jan 2025 – Present

Developer: Created a real-time EEG-driven binaural beat system using reinforcement learning to optimize frequency and phase shifts to promote specific mental states

- **Real-Time EEG Processing:** Built a WebSocket pipeline using Python and asyncio for low-latency brainwave data acquisition.
- **Machine Learning & Signal Processing:** Applied reinforcement learning to optimize binaural beat frequency and phase shifts based on EEG responses.

Cooperative Virtual Reality Game, Austin, TX

Aug 2023 – Dec 2023

Developer: Worked with a team to develop a cooperative virtual reality (VR) game.

- **VR Game Development:** Developed immersive mechanics and gameplay in Unity using C# with a focus on collaboration, resource management, and combat systems.
- **Graphics & Physics:** Designed an interactive crafting system with modular weapon components, leveraging computer graphics techniques.

VirtualMouse AI, Austin, TX

Sep 2023 – Sep 2023

Developer: developed a real-time hand tracker, gesture recognition, and control system.

- **Computer Vision & AI:** Built a real-time hand tracking and gesture recognition system using Python, OpenCV, and Google's Mediapipe.